Diffeomorphic Fourier Neural Operators for Zero-Shot Physics-Informed Learning on Complex Geometries

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***Abstract*—Fourier Neural Operators (FNOs) struggle with arbitrary non-rectangular domains due to spectral gibbs phenomena and grid representation limits. We propose DIF-FNO, a diffeomorphic operator framework mapping physical domains to standard latent domains. By enforcing positive Jacobian determinants $\det(J) > 0$ and evaluating physical derivatives via the geometric Chain Rule, DIF-FNO achieves resolution-invariant zero-shot generalization with under 1% relative L^2 error.**

***Index Terms*—Fourier Neural Operator, Diffeomorphism, Physics-Informed Neural Networks, Partial Differential Equations.**

I. Introduction

Deep learning models for operator learning, particularly Fourier Neural Operators (FNO), have demonstrated remarkable speedups over traditional numerical solvers. However, their reliance on regular Cartesian grids presents severe challenges when applied to irregular or parametric physical geometries.

II. Methodology

A. *Diffeomorphic Latent Mapping*

Let $\Omega_p \subset \mathbb{R}^2$ be the physical domain and $\Omega_l = [-1, 1]^2$ be the reference latent domain. We define a smooth bi-differentiable mapping $\phi : \Omega_l \to \Omega_p$ such that $(x, y) = \phi(\xi, \eta)$. To eliminate grid folding and preserve topological validity, we enforce the Jacobian determinant condition:

$$\det(J) = \frac{\partial x}{\partial \xi}\frac{\partial y}{\partial \eta} - \frac{\partial x}{\partial \eta}\frac{\partial y}{\partial \xi} > 0 \quad \forall(\xi, \eta) \in \Omega_l \tag{1}$$

B. *Physical Derivative Calculation via Chain Rule*

To accurately evaluate gradient norms in physical space, latent derivatives $\nabla_{(\xi,\eta)} u$ are transformed using the inverse transposed Jacobian:

$$\begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \end{bmatrix} = J^{-T} \begin{bmatrix} \frac{\partial u}{\partial \xi} \\ \frac{\partial u}{\partial \eta} \end{bmatrix} = \frac{1}{\det(J)} \begin{bmatrix} \frac{\partial y}{\partial \eta} & -\frac{\partial y}{\partial \xi} \\ -\frac{\partial x}{\partial \eta} & \frac{\partial x}{\partial \xi} \end{bmatrix} \begin{bmatrix} \frac{\partial u}{\partial \xi} \\ \frac{\partial u}{\partial \eta} \end{bmatrix} \tag{2}$$

III. Experimental Results

We benchmark DIF-FNO against Geo-FNO and FNO-Mask across unseen parametric polar domains.

As detailed in Table **??**, DIF-FNO consistently outperforms baseline methods, maintaining high accuracy across grid resolution scaling.

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TABLE I

Deformed Darcy Flow Benchmark: Convergence Results

	64×64		128×128	
Model	L^2 Error	H^1 Error	L^2 Error	H^1
DIF-FNO (Ours)	0.0541 $\pm$0.0031	0.3173 $\pm$0.0097	0.0617 $\pm$0.0023	0.3679 $\pm$
Geo-FNO	0.0645 $\pm$0.0046	0.2991 $\pm$0.0047	0.0705 $\pm$0.0039	0.3418 $\pm$
Masked-FNO	0.1042 $\pm$0.0088	0.2986 $\pm$0.0040	0.1074 $\pm$0.0087	0.3243 $\pm$

IV. Conclusion

DIF-FNO provides a mathematically rigorous framework for solving PDEs on complex domains while guaranteeing topological consistency.